

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment**

|                |                                                                                                                                                 |               |                                                    |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------------------------------------|
| Course Code:   | CSA10 / CSA1063                                                                                                                                 | Course Title: | Software Engineering                               |
| CO Assessed:   | CO2                                                                                                                                             | Assessment:   | Assessment Tool 2 – Architecture Design Assignment |
| Weightage:     | 20%                                                                                                                                             | Total Marks:  | 25                                                 |
| CO2 Statement: | <i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i> |               |                                                    |
| Student Name:  | RAVURU PRANATHI                                                                                                                                 | Reg. No.:     | 192525076                                          |
| Date:          | 03-08-2026                                                                                                                                      | Duration:     | 60 minutes                                         |

**Code of Conduct**

*I R.Pranathi(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: 

**Annexure A – Architecture Design Assignment**

**Instructions to Students:**

Each student will be assigned an **individual software project problem statement**. Based on the assigned problem, analyze the system requirements and design an appropriate software architecture by applying software engineering principles. Your submission should demonstrate your ability to select, justify, and evaluate a suitable architectural style.

**Question**

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

**1. Analyze System Requirements**

- Study the given problem statement and identify the system objectives.
- Analyze the functional and non-functional requirements that influence the software architecture.
- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

**2. Select a Suitable Software Architecture**

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC, Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).
- Justify why the selected architecture is the most suitable for the given problem.

**3. Draw the Software Architecture Diagram**

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.
- Use appropriate software architecture notations and labels.

**4. Explain Components and Their Interactions**

- Describe the purpose and responsibilities of each architectural component.

- Explain how the components communicate and exchange data.
- Illustrate the overall workflow of the system.

## 5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- Security
- Performance
- Reliability
- Maintainability
- Availability

## 6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
- Discuss the trade-offs considered while selecting the architecture.
- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

## Rubrics for Calculation

| Evaluation Criteria                              | Excellent (4 Marks)                                                                                                                                                                | Good (3 Marks)                                                          | Satisfactory (2 Marks)                                                  | Needs Improvement (1 Mark)                                        |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>System Requirement Analysis</b>               | Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture. | Analyzes most requirements with minor omissions or limited explanation. | Identifies only basic requirements with partial analysis.               | Requirement analysis is incomplete, inaccurate, or lacks clarity. |
| <b>Selection of Software Architecture</b>        | Selects the most appropriate architectural style with strong technical justification aligned to system requirements.                                                               | Selects a suitable architecture with reasonable justification.          | Architecture is partially suitable with limited justification.          | Inappropriate architecture selected or justification is missing.  |
| <b>Architecture Diagram and Component Design</b> | Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.                               | Diagram is mostly complete with minor omissions or notation issues.     | Diagram includes only essential components and lacks sufficient detail. | Diagram is incomplete, unclear, or technically incorrect.         |

|                                                               |                                                                                                                                                                         |                                                                            |                                                                                                        |                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Component Interaction and Quality Attribute Analysis</b>   | Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability. | Explains most components and quality attributes with minor gaps.           | Provides limited explanation of interactions and addresses only some quality attributes.               | Component interactions and quality attributes are poorly explained or missing.    |
| <b>Architectural Justification and Technical Presentation</b> | Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and presents a well-organized, professional report.            | Provides adequate justification with minor gaps; report is well organized. | Limited justification with minimal discussion of trade-offs or future scope; report needs improvement. | Justification is weak or absent; report lacks organization and technical clarity. |

| Criteria                                               | Mark      |
|--------------------------------------------------------|-----------|
| System Requirement Analysis                            | /5        |
| Selection of Software Architecture                     | /5        |
| Architecture Diagram and Component Design              | /5        |
| Component Interaction and Quality Attribute Analysis   | /5        |
| Architectural Justification and Technical Presentation | /5        |
| <b>TOTAL</b>                                           | <b>/5</b> |

## **1. System Requirement Analysis**

### **1.1 Project Title**

#### **Smart Construction Project Management System**

### **1.2 Introduction**

The **Smart Construction Project Management System (SCPMS)** is a cloud-based software system designed to manage construction projects efficiently through a centralized platform. It enables project managers, engineers, contractors, workers, and administrators to coordinate construction activities, monitor project progress, manage resources, store documents, assign tasks, and receive real-time notifications. The system reduces manual work and improves communication by integrating project management, task tracking, resource management, document management, reporting, and notification services into a single system.

### **1.3 System Objectives**

The main objectives of the system are:

1. To provide centralized management of construction projects.
2. To allow project managers to create, update, and monitor projects.
3. To assign construction tasks to engineers, contractors, and workers.
4. To track task progress, deadlines, and completion status.
5. To manage construction resources such as materials, equipment, and manpower.
6. To provide secure storage and management of project documents.
7. To generate project progress reports and dashboards.
8. To provide real-time notifications for important project activities.
9. To improve communication and collaboration among stakeholders.
10. To provide a scalable and reliable platform for multiple construction projects.

### **1.4 Functional Requirements**

#### **FR1 – User Registration and Login**

The system shall allow users to register, log in, and securely access the system.

#### **FR2 – Role Management**

The system shall support different roles such as:

- Administrator
- Project Manager
- Site Engineer
- Contractor
- Worker

Each role will have different access permissions.

#### **FR3 – Project Management**

The project manager shall be able to:

- Create projects
- Edit project details
- Assign project members
- Set project deadlines
- Monitor project status
- Close completed projects

#### **FR4 – Task Management**

The system shall allow users to:

- Create tasks
- Assign tasks
- Set priorities
- Set deadlines
- Update task status
- Monitor task completion

#### **FR5 – Resource Management**

The system shall manage:

- Construction materials
- Equipment
- Labour/manpower
- Resource availability
- Resource allocation

#### **FR6 – Document Management**

Users shall be able to upload and manage:

- Construction plans
- Design documents
- Project reports
- Contracts
- Site photographs
- Inspection documents

#### **FR7 – Progress Tracking**

The system shall provide real-time information about project and task progress.

#### **FR8 – Notifications**

The system shall notify users about:

- New task assignments
- Upcoming deadlines
- Delayed tasks
- Project updates
- Important announcements

#### **FR9 – Reporting**

The system shall generate reports related to:

- Project progress
- Task completion
- Resource usage
- Delayed activities
- Project performance

#### **FR10 – Dashboard**

The system shall provide dashboards showing important project information in a summarized form.

### **1.5 Non-Functional Requirements**

The assignment specifically requires analysis of quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

## **1. Scalability**

The system should support increasing numbers of users, projects, tasks, and documents without major performance degradation.

## **2. Security**

The system should provide:

- Secure authentication
- Role-based access control
- Data encryption
- Secure API communication
- Protected document access

## **3. Performance**

The system should provide fast response times for:

- Dashboard loading
- Task updates
- Project searches
- Report generation
- Data retrieval

## **4. Reliability**

The system should continue functioning correctly even when individual services encounter errors.

## **5. Maintainability**

The architecture should be modular so that individual components can be modified or upgraded without affecting the entire system.

## **6. Availability**

The system should be accessible 24/7 with minimal downtime through cloud deployment, backup, and recovery mechanisms.

## **2. Selection of Suitable Software Architecture**

### **Selected Architecture: Layered / N-Tier Architecture**

For the **Smart Construction Project Management System**, the most suitable architecture is a **Layered (N-Tier) Architecture**.

The assignment allows architectural styles such as Layered, Client-Server, MVC, Microservices, Event-Driven, SOA, Serverless, and Hybrid architecture, and requires justification for the selected style.

### **2.1 Proposed Layers**

The system consists of:

1. **Presentation Layer**
2. **Application Layer**
3. **Business Logic Layer**
4. **Data Access Layer**
5. **Database Layer**

External services such as email, SMS, cloud storage, and payment services can communicate through APIs.

### **2.2 Why Layered Architecture is Suitable**

Layered architecture is suitable because the construction management system contains several logically separate responsibilities.

#### **Separation of Concerns**

Each layer performs a specific responsibility.

## Modularity

New features such as AI-based delay prediction can be added without redesigning the complete system.

## Security

Authentication and authorization can be implemented between the presentation, application, and business layers.

## Maintainability

Developers can modify one layer without significantly affecting other layers.

## Scalability

The application and database components can be scaled according to workload.

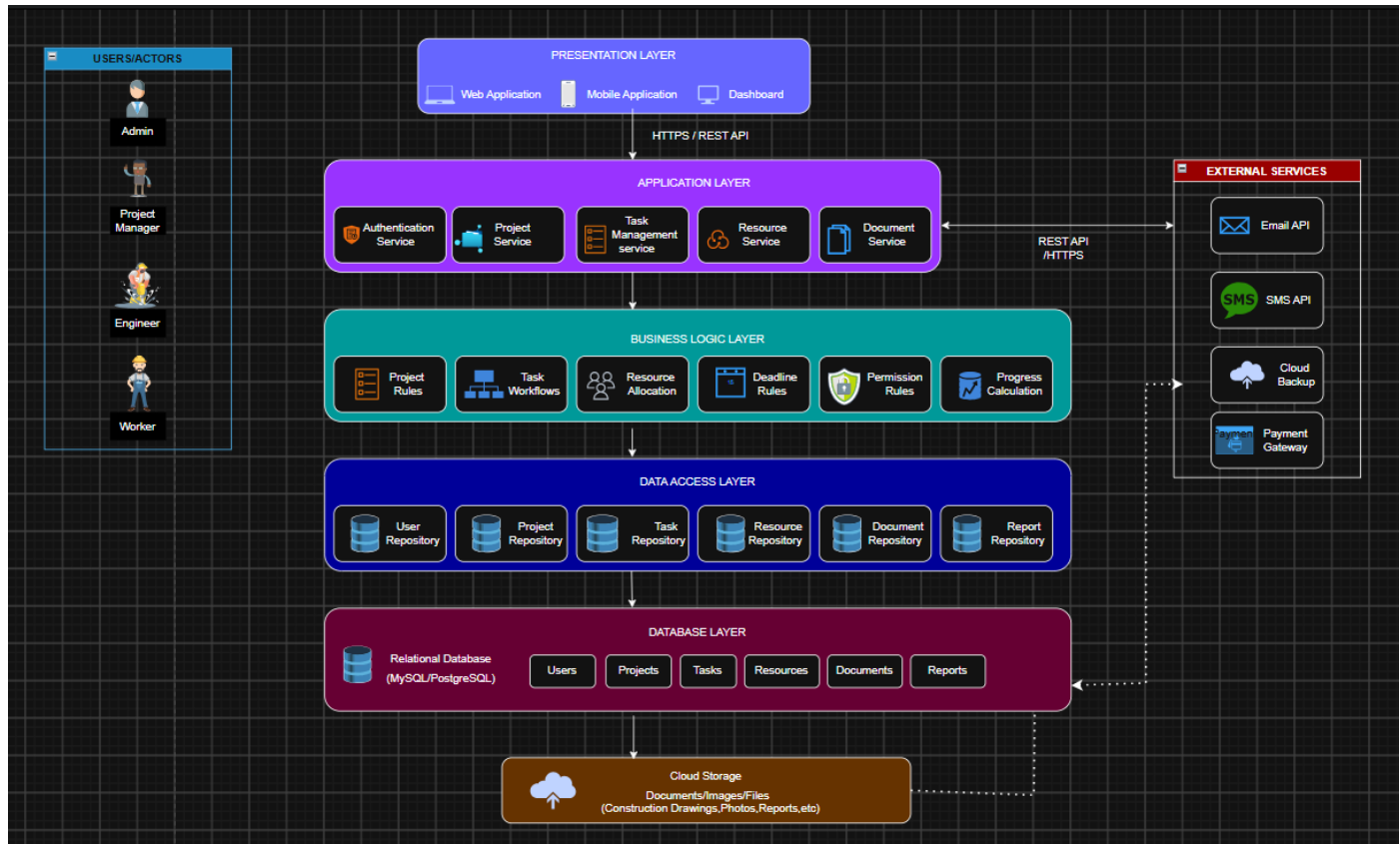
## Integration

The system can communicate with external services such as:

- Email APIs
- SMS APIs
- Cloud storage
- Payment gateways
- Mapping/location services

## 3. Software Architecture Diagram

Your assignment requires the architecture diagram to show **major layers/components, databases, external systems, APIs, and communication mechanisms.**



## Communication Mechanisms

Use these labels on your actual diagram:

- **HTTPS** – secure communication between users and system.
- **REST API** – communication between frontend and backend.
- **SQL/ORM** – communication between application and database.
- **Cloud API** – document and file storage.
- **Email/SMS API** – notification delivery.
- **Authentication Token/JWT** – secure user authentication.

## 4. Components and Their Interactions

The assignment expects both **component responsibilities** and **how components exchange data**, along with the overall workflow.

### 4.1 Presentation Layer

The presentation layer provides the user interface.

#### Responsibilities:

- Login and registration screens
- Project dashboard
- Task management interface
- Resource management interface
- Document upload interface
- Reports and charts
- Notifications

Users access the system through a web browser or mobile application.

### 4.2 Authentication Service

The authentication service manages:

- User login
- User registration
- Password validation
- Session/token management
- Role-based access control

For example, an administrator can manage users, while a worker may only view and update assigned tasks.

### 4.3 Project Management Service

This component manages construction projects.

It performs:

- Project creation
- Project modification
- Project assignment
- Project status tracking
- Project deadline management

### 4.4 Task Management Service

This component manages construction activities.

It handles:

- Task creation
- Task assignment
- Task priority
- Task deadlines
- Task status
- Task completion

Example:

Project Manager



Creates Task



Assigns Engineer/Worker



Worker Updates Status



System Updates Project Progress

#### **4.5 Resource Management Service**

This component manages:

- Materials
- Equipment
- Workers
- Resource availability
- Resource allocation

It helps project managers identify whether required resources are available.

#### **4.6 Document Management Service**

The document service handles:

- Uploading documents
- Downloading documents
- Document categorization
- Version management
- Secure access

Large files can be stored in cloud storage while metadata is stored in the database.

#### **4.7 Notification Service**

The notification service sends alerts for:

- New assignments
- Deadline reminders
- Delayed tasks
- Project updates
- Important announcements

It can communicate with external Email and SMS APIs.

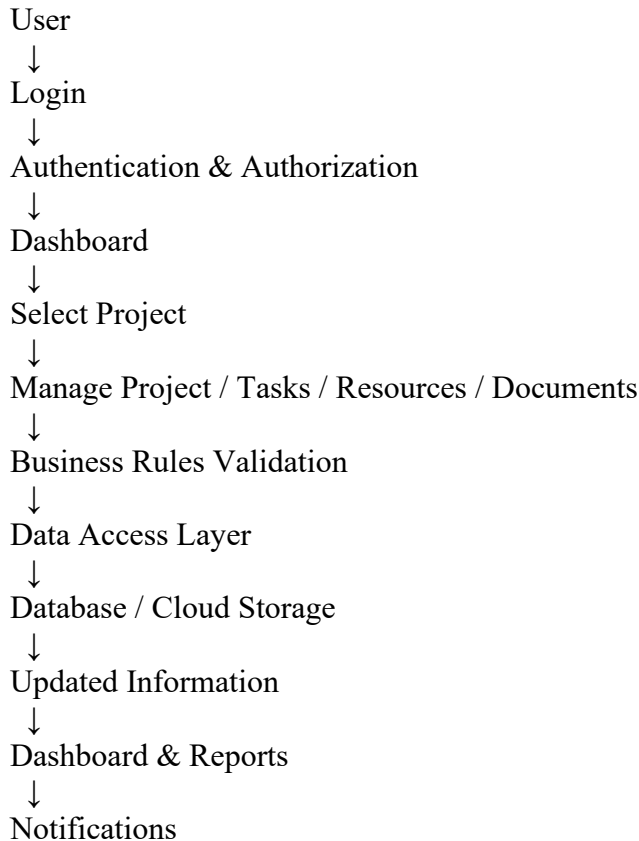
#### **4.8 Reporting Service**

The reporting service collects information from projects, tasks, and resources and generates:

- Project progress reports
- Task completion reports
- Resource utilization reports
- Delay reports
- Performance reports

## 4.9 Overall System Workflow

The system workflow is:



## 5. Quality Attribute Analysis

The rubric gives specific importance to **scalability, security, performance, reliability, maintainability, and availability**.

| Quality Attribute      | Architectural Solution                                                                     |
|------------------------|--------------------------------------------------------------------------------------------|
| <b>Scalability</b>     | Application servers can be scaled when the number of users/projects increases.             |
| <b>Security</b>        | Authentication, role-based access control, HTTPS, encryption and secure APIs protect data. |
| <b>Performance</b>     | Database indexing, optimized queries, caching and efficient APIs improve response time.    |
| <b>Reliability</b>     | Transactions, error handling, backups and recovery mechanisms reduce data loss.            |
| <b>Maintainability</b> | Layer separation allows individual modules to be modified independently.                   |
| <b>Availability</b>    | Cloud deployment, backup and redundant services support continuous availability.           |

### Scalability

The architecture allows application components to be scaled independently. For example, if many users access the task management module, additional application resources can be provided without changing the entire system.

### Security

Security is implemented through:

- Role-based access control
- Secure authentication
- HTTPS

- Password hashing
- Authorization checks
- Database access controls
- Secure document permissions

## **Performance**

Performance can be improved through:

- Database indexing
- Query optimization
- Caching
- Efficient REST APIs
- Load balancing
- Asynchronous notifications

## **Reliability**

Reliability is achieved using:

- Database transactions
- Automatic backups
- Error handling
- Data validation
- Recovery mechanisms

## **Maintainability**

The layered structure separates presentation, business logic, and data access. Therefore, developers can modify a particular component without redesigning the complete system.

## **Availability**

Cloud deployment with backups and redundant services helps the system remain available to construction teams throughout the day.

## **6. Architectural Decisions and Justification**

The assignment requires the architectural choices to be justified, including **trade-offs, future enhancements, system integration, and long-term maintainability**.

### **6.1 Why Layered Architecture Was Selected**

Layered architecture was selected because the system contains multiple responsibilities that can be clearly separated.

The presentation layer handles user interaction, the application layer manages application services, the business layer handles construction-related rules, the data access layer manages database operations, and the database layer stores information.

This separation improves:

- Modularity
- Security
- Maintainability
- Scalability
- Testing
- Future development

### **6.2 Architectural Trade-Offs**

#### **Advantages**

- Simple and structured design
- Easy to understand
- Easy maintenance

- Clear separation of responsibilities
- Good security control
- Easier testing
- Supports future expansion

### **Disadvantages**

- Additional layers can increase processing overhead.
- Large systems may require more infrastructure.
- Poorly designed layers can create unnecessary dependencies.

### **Decision**

The benefits of layered architecture are greater than its limitations because the proposed construction management system requires **security, maintainability, scalability, and clear separation of responsibilities**.

### **6.3 Future Enhancements**

The architecture can support future features such as:

1. AI-based construction delay prediction
2. IoT sensor integration
3. Drone-based site monitoring
4. Computer vision for safety monitoring
5. AI-based resource optimization
6. Predictive maintenance
7. Real-time GPS tracking
8. Advanced analytics dashboards
9. Digital twin integration
10. Automated construction progress estimation

These features can be added as new services without completely redesigning the existing architecture.

## **7. Final Architectural Summary**

The **Smart Construction Project Management System** uses a **Layered/N-Tier Architecture** consisting of the Presentation, Application, Business Logic, Data Access, and Database layers.

The architecture provides a clear separation of responsibilities and supports secure communication between users, application services, databases, cloud storage, and external APIs.

The architecture satisfies the major requirements of the system by providing:

- **Scalability** for multiple construction projects
- **Security** through authentication and access control
- **Performance** through optimized data access
- **Reliability** through backups and error handling
- **Maintainability** through modular layers
- **Availability** through cloud-based deployment

Therefore, the proposed architecture is suitable for a modern construction management platform and provides a strong foundation for future integration of **AI, IoT, drones, analytics, and smart construction technologies**.